MUHAMMAD IMRAN	BT201 Ecology, Biodiversity & Evolution-II
Past papers solved QUESTIONS	MID TERM

1. How gene can be identified?

Disease **gene identification** is a process by which scientists **identify** the mutant genotypes responsible for an inherited **genetic** disorder. Mutations in these **genes can** include single nucleotide substitutions, single nucleotide additions/deletions, deletion of the entire **gene**, and other **genetic** abnormalities. **(Internet)**

In a fully sequenced genome, genes are found by scanning the sequence using gene-predicting computer programmes and assigning putative functions by searching for similarities in already existing databases. **(PPTS)**

2. Three point of phylogenetic? (Domain)

Bacteria, Archaea and Eukarya

3. What is bionomenclature?

In the 18th century, Carolus Linnaeus published a system of taxonomy based on resemblances. Two key features of his system remain useful today: two-part names for species and hierarchical classification. The two-part scientific name of a species is called a binomial. The first part of the name is the genus. The second part, called the specific epithet, is unique for each species within the genus. The first letter of the genus is capitalized, and the entire species name is italicized or underlined. Both parts together name the species. **(PPTS)**

Binomial nomenclature, also called binominal nomenclature or binary nomenclature, is a formal system of naming species of living things by giving each a name composed of two parts, both of which use Latin grammatical forms, although they can be based on words from other languages. **(Internet)**

4. Micro ecology?

Microecology means microbial ecology or ecology of a microhabitat. Human gut **microecology** is the study of microbial ecology of the human gut. **Microecology** is a large field which includes many topics such as evolution, biodiversity, exobiology, ecology, bioremediation, recycling, and food microbiology.

5. Species diversity? Species Richness?

Species diversity is defined as the number of **species** and abundance of each **species** that live in a particular location. The number of **species** that live in a certain location is called **species** richness.

6. Different in ortholog and paralog gene?

Orthologs are **genes** that are related by vertical descent from a common ancestor and encode proteins with the same function in **different** species.
Paralogs are homologous **genes** that have evolved by duplication and code for protein with similar, but not identical functions."

7. Importance of bacteria and fungi?

Fungi and **bacteria** are essential to many of the most basic ecosystem processes: saprobic **fungi** break down fallen wood and litter returning nutrients to the soil; nitrogen-fixing **bacteria** and mycorrhizal **fungi** assist plants to obtain nutrients from the soil; and many groups of **fungi** and **bacteria** cause diseases of plants

8. History of plant breeding?

Gregor Mendel's experiments with plant hybridization led to his establishing laws of inheritance. Once this work became well known, it formed the basis of the new science of genetics, which stimulated research by many plant scientists dedicated to improving crop production through plant breeding

9. Bioinformatics?

Bioinformatics combines biology, computer science, information engineering, mathematics and statistics to analyze and interpret biological data. **Bioinformatics** has been used for in silico analyses of biological queries using mathematical and statistical techniques

10. Lack of fossil record?

At the time of Darwin, the fossil record was VERY scanty. A great deal of progress has been made since, but it is far from finished.

The fossil record is relatively incomplete for several reasons: 1. Soft tissues are rarely preserved 2. Movement of the earth's crust has obliterated and/or covered many fossils 3. Fossilization takes place only in certain types of habitats and favorable environments 4. Paleontologists have not dug up every place on earth

11. Enlist model?

Some models • Dynamic programming • Hidden Markov Models (HMMs) • Conditional random field (CRF) • Support vector machines (SVMs)

12. Evolution of organism documents from phylogeny? How evolutionary history convert into documents?

- Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history
- DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago
- mtDNA evolves rapidly and can be used to explore recent evolutionary events

13. Name phyla in amoeboid protozoa and their locomotion?

Protozoa can be divided into four **phyla** based on **their locomotion**: Mastigophora, Sarcodina, Ciliophora, and Sporozoa.

14. Systematics?

systematics (cladistics) is a method of taxonomic classification based on their evolutionary history. It was developed by Willi Hennig, a German entomologist, in 1950.

15. Species richness Systematics?

Species richness is the number of different species in a particular community. If we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first.

16. Phylogeny?

Phylogeny is the evolutionary history of a species or group of related species The discipline of systematics classifies organisms and determines their evolutionary relationships

17. Difference between arctic and polar region?

the **Arctic is** not a continent which **is** the predominant different **between** the two **polar regions**. Under the massive ice cap **of** the **Arctic** lies the **Arctic Ocean** whose depths reach 15,000 feet below the surface. Often times the two **polar regions** are confused due to their having such similar names. **(Internet)**

18. Law of succession?

General correspondence between fossil and living forms in same geographical area • Darwin used this observation to predict that fossils of ancient humans should be found in Africa

19. Why breeding crop important?

Crop breeding is the art and science of improving **important** agricultural plants for the benefit of humankind. **Crop breeders** work to make our food, fiber, forage, and industrial **crops** more productive and nutritious. **Crops** provide for an expanding global population with increasing dietary expectations. **(Internet)**

20. Factor affect commute structure?

The gender, age, number of family cars, housing source and jobs-housing balance are the most important **factors** influencing **commuting** mode choice; housing, population density, jobs-housing balance and **commuting** mode significantly **affect commuting** time; and transport accessibility, jobs-housing balance and **commuting** mode **(Internet)**

21. Ecology?

Ecology is a branch of biology that studies the interactions among organisms and their biophysical environment, which includes both biotic and abiotic components.

"Ecology is the scientific study of the processes regulating the distribution and abundance of organisms and the interactions among them, and the study of how these organisms in turn mediate the transport and transformation of energy and matter in the biosphere (i.e., the study of the design of ecosystem structure and function).

22. Applied Ecology?

Applied Ecology: Using ecological principles to maintain conditions necessary for the continuation of present day life on earth.

23. Plants breeding?

Plant breeding is the art and science of changing the traits of plants in order to produce desired characteristics. Plant breeding can be accomplished through many different techniques ranging from simply selecting plants with desirable characteristics for propagation, to methods that make use of knowledge of genetics and chromosomes, to more complex molecular techniques.

24. Protozoa life cycle?

Protozoa have two-phase **life cycles**, alternating between proliferative stages (e.g., trophozoites) and dormant cysts. As cysts, **protozoa** can survive harsh conditions, such as exposure to extreme temperatures or harmful chemicals, or long periods without access to nutrients, water, or oxygen for periods of time. **(Internet)**

25. Five scientist citing reteach?

- **Kerner** showed that the differences in growth form observed by him and Bonnier were the result of temporary adjustments to environmental variation and not hereditary changes in the experimental plants.
- **Turesson's** studies provided evidence for genetic differences among populations.
- **Clausen, Keck, and Hiesey** explored the extent and sources of morphological variation in plant populations, including both the influences of environment and genetics.
- **Case** determined that the best predictor of chuckwalla, Sauromalus, body length was average winter rainfall
  - **Tracy's** laboratory growth experiments indicated that variation in body size among chuckwalla populations is at least partly determined by genetic differences among populations.

26. How climate affect community?

Climate change could **affect** our **society** through impacts on a number of different social, cultural, and natural resources. For example, **climate** change could **affect** human health, infrastructure, and transportation systems, as well as energy, food, and water supplies. **(Internet)**

27. Sequencing?

DNA **sequencing** is the process of determining the nucleic acid sequence – the order of nucleotides in DNA. It includes any method or technology that is used to determine the order of the four bases: adenine, guanine, cytosine, and thymine.

Long 5 Marks

28. Evolutionary related study to genome sequence?

Explore genome evolution based on large data sets of DNA or protein sequences. Using entire genomes to infer a species tree (Eisen and Fraser 2003). Based on maximum genetic information and average out the anomalies. Has become the standard for reconstructing reliable phylogenies (Ciccarelli et al, 2006; Daubin et al. 2002).

29. Plant breeding?

Plant breeding is the science of changing the traits of **plants** in order to produce desired characteristics. It has been used to improve the quality of nutrition in products for humans and animals. ... **Plant breeding** has been practiced for thousands of years, since near the beginning of human civilization.

30. Why we use phylogenic tree? Why phylogenic tree is important? Reasons to use phylogenetic tree?

Phylogenetics is important because it enriches our understanding of how genes, genomes, species (and molecular sequences more generally) evolve.

Phylogenetics is **important** because it enriches our understanding of how genes, genomes, species (and molecular sequences more generally) evolve. **(Internet)**

31. Biodiversity importance in term of food and human?

Biodiversity creates resilience and is a key to mitigate the risks in agriculture particularly farming, from where we get most of our **food**. It ensures the world gets nutrients which is critical for good **health** and is also critical for conserving the ecosystem that produces the **food**. **(Internet)**

32. Factor forming community shape?

Shape The Forces That Bring A Community Together

- Common goals. Common goals (whether long-term or short-term) unite members to contribute towards it. ...
- Common threat. Common threats, problems and enemies help forge groups together. ...

- Isolation from others. Isolation from others focuses the attention upon each other. ... Barriers to entry. Barriers to entry are important. **(Internet)**

33. Landscape ecology explain?

Landscape ecology is the science of studying and improving relationships between **ecological** processes in the environment and particular ecosystems. This is done within a variety of **landscape** scales, development spatial patterns, and organizational levels of research and policy. **(Internet)**

34. Linnaeus classification?

Carolus Linnaeus is the father of taxonomy, which is the system of classifying and naming organisms. One of his contributions was the development of a hierarchical system of classification of nature. Today, this system includes eight taxa: domain, kingdom, phylum, class, order, family, **genus**, and **species** **(Internet)**

35. Phylum Annelida?

Annelid, **phylum** name **Annelida**, also called segmented worm, any member of a **phylum** of invertebrate animals that are characterized by the possession of a body cavity (or coelom), movable bristles (or setae), and a body divided into segments by transverse rings, or annulations, from which they take their name. **(Internet)**

Phylum Annelida Class Polychaeta Family Siboglinidae Class Clitellata • Subclass Oligochaeta. Subclass Hirudinea

Annelid Characteristics • Defining Characteristics • One or more pairs of chitinous setae • The phylum includes polychaetes, earthworms, leeches, and vestimentiferans • True segmented worms • Metameric segmentation

Body Structure The body is a tube within a tube. The coelom is important to annelids for:The epidermis is what secretes the tough cuticle

Locomotion On each side of the animal is a parapod (parapodia) consisting of fleshly lobes, which are supported by chitinous rods. Each parapod have setae, which can be sharp (protection), and aid in locomotion

Feeding Annelids range from carnivores, herbivores, scavengers, deposit feeders, and filter feeders. With very few defenses, many remain in a burrow or secreted tube. Carnivores can capture prey with strong jaws and quickly drag it back to its burrow. Can use a muscular pharynx = eversible proboscis. **(PPTS)**

36. Define community structure? 2 marks

Community structure is essentially the composition of a community, including the number of species in that community and their relative numbers. It can also be interpreted more broadly, to include all of the patterns of interaction between these different Species.

37. what is intermediate distribution hypothesis. (2)

The intermediate disturbance hypothesis suggests that communities with a medium (intermediate) level of disturbance may have greater species diversity than communities with very frequent or very rare disturbances.

38. Ecological genomics..2

A scientific discipline that studies the structure and functioning of a genome with the aim of understanding the relationship between the organism and its biotic and abiotic environments

39. Enlist four reasons for lacking of fossil record?2 marks

1. Soft tissues are rarely preserved
2. Movement of the earth's crust has obliterated and/or covered many fossils
3. Fossilization takes place only in certain types of habitats and favorable environments
4. Paleontologists have not dug up every place on earth

40. Explain species richness and species diversity with example? 3 marks

Species richness is the number of different species in a particular community. If we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first. Communities with the highest species richness tend to be found in areas near the equator, which have lots of solar energy (supporting high primary productivity), warm temperatures, large amounts of rainfall, and little seasonal change. Communities with the lowest species richness lie near the poles, which get less solar energy and are colder, drier, and less amenable to life

Species diversity is a measure of community complexity. It is a function of both the number of different species in the community (species richness) and their relative abundances (species evenness). Larger numbers of species and more even abundances of species lead to higher species diversity. For example: • A forest community with 20 different kinds of trees would have greater species diversity than a forest community with only 5 kinds of trees (assuming that the tree species were even in abundance in both cases)

41. Explain importance of biodiversity? 5 marks

These Are As Follows:

a. Maintaining Balance of The Ecosystem.
b. Recycling And Storage of Nutrients,
c. Combating Pollution, And Stabilizing Climate, Protecting Water Resources,
d. Forming And Protecting Soil And Maintaining Eco-Balance.
e. Intrinsic Value
f. A Philosophical Concept of The Value of Something Independent of Its Value To Anyone or Anything

42. What are phylogenetic trees (How genes are related to each other) ? 5 marks

Explore genome evolution based on large data sets of DNA or protein sequences.
Using entire genomes to infer a species tree (Eisen and Fraser 2003). Based on maximum gene

43. Factors affacting comunity structure

a. The climate patterns of the community's location.
b. The geography of the community's location
c. The heterogeneity (patchiness) of the environment
d. The frequency of disturbances, or disruptive events
e. Interactions between organisms

44. Enlist three major technological developments of 1990s.

MICROECOLOGY • COMPUTING • COMMUNICATION.

45. Enlist types of servers used in bioinformatics.

WWW servers
Database servers
Intensive computing servers

46. How sequence assembly is done using shotgun approach?

DNA is broken up randomly into numerous small segments, which are sequenced using the chain termination method to obtain reads. Multiple overlapping reads for the target DNA are obtained by performing several rounds of this fragmentation and sequencing. Computer programs then use the overlapping ends of different reads to assemble them into a continuous sequence.[

47. What phyla include amoeboid protozoan..Write their mode of locomotion...

Amoebozoa

It is a crawling-like type of movement accomplished by protrusion of cytoplasm of the cell involving the formation of pseudopodia

48. How the evolutionary history convert in to document?

- Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history
- DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago
- mtDNA evolves rapidly and can be used to explore recent evolutionary events

49. Write at least three reasons to use phylogenetic.

Understand the lineage of different species
Organizing principle to sort species into taxonomy
Understand how various functions evolved
Understand forces and constraints on evolution
Perform multiple sequence alignment

Predict gene function (phylogenetic footprint)

50. Differentiate between species tree and gene tree.

Species tree (how are my species related?) contains only one representative from each species when did speciation take place? all nodes indicate speciation events **Gene tree** (how are my genes related?) often contains a number of genes from a single species. nodes relate either to speciation or gene duplication events.

51. Define genome

A genome is an organism's complete set of DNA, including all of its genes. Each genome contains all of the information needed to build and maintain that organism.

52. What does intermediate disturbance hypothesis suggest? (2 marks)

Ans: The intermediate disturbance hypothesis suggests that communities with a medium (intermediate) level of disturbance may have greater species diversity than communities with very frequent or very rare disturbances.

53. interelation ship between gene, genome and dynamic character of gene.

Ans: The interactions between the genes within the genome and the dynamic character of the genome
• sketched by Dover (1999) as an internal tangled bank. • idea of Darwin (1859) • Dover (1999 :
– 'external tangled bank' (the ecology) – the 'internal tangled bank' (the genome) the internal tangled bank

- It emphasizes the role of genetic turbulence (gene duplication, genetic sweeps, exon shuffling, transposition, etc.)
- illustrates that there is ample scope for 'innovation from within genetic turbulence

The genomics revolution

- Three major technological developments of 1990s

– microtechnology – computing – communication.
(note :further details in topic 3)

54. what is primer in PCR?. (3 marks)

Ans: Attempts to pick up the gene by polymerase chain reaction (PCR) using primers from the then-known drosophila mt sequence were unsuccessful, which was

55. Write at least five factors that shape community structure. (5 marks)

Ans: The climate patterns of the community's location.

- Global climate patterns (variation in temperature, solar energy input, etc. With distance from the equator) can affect community structure. So can more local climate patterns, such effects created by mountain ranges, bodies of water, and even streams and valleys.

- The predictability or variability of climate can also affect community structure – for instance, some species may be unable to survive in a region with sporadic droughts or sporadic drops below freezing.

The geography of the community's location.

Geographical features of the area in which the community is found can affect community structure. For instance, island communities that are further away from a mainland tend to have smaller numbers of species than those that are closer to the mainland. This reflects that the chance event of a species arriving from the mainland is less likely when the island is more distant.

The heterogeneity (patchiness) of the environment

- If there is more variation, or heterogeneity, in a community's environment, this may allow for greater species richness because there are more distinct habitats to be occupied.

- For example, imagine a community occupying a field, and another occupying a field that is dotted with piles of rock. The second community may have greater species richness because species that can live in the rocks (but not in the open field) will be present, in addition to those species that can live in the field.

The frequency of disturbances, or disruptive events.

- The frequency of disruptive events (including storms, wildfires, and landslides) can affect the structure of a community. The intermediate disturbance hypothesis suggests that communities with a medium (intermediate) level of disturbance may have greater species diversity than communities with very frequent or very rare disturbances.

Interactions between organisms.

- All of the interspecies interactions that can occur between organisms (competition, predation, and various forms of symbiosis) have the potential to shape a community. For instance, two species that compete intensively with one another may be unable to coexist in the same community, or a prey species may be unable to persist in a community that contains a highly effective predator.

56. Define bioinformatics. (2 marks)

The field of biology specializing in developing hardware and software to store and analyze the huge amounts of data being generated by life scientists.
(NIH)
Key Challenge of Bioinformatics
The world of biology is very different from what it was even ten years ago. To bridge the considerable gap between technical data production and its use by scientists for biological discovery

57. Differentiate between species richness and species diversity with examples. (3 marks)

richness
Species richness is the number of different species in a particular community. If we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first. Diversity
Species diversity is a measure of community complexity. It is a function of both the number of different species in the community (species richness) and their relative abundances (species evenness). Larger numbers of species and more even abundances of species lead to higher species diversity.
(note: further details in topic 23)

58. Write names of phyla of ameboid protozoa. Also write their mode of locomotion. (3 marks)

Contains 4 phyla: Foraminifera, Radiozoa, Amoebozoa, and Heliozoa

- Most reproduce asexually through binary fission
- Characterized by pseudopodia
- Food is usually captured by phagocytosis
- Body types range from free flowing to rigid with skeletal supports

59. Significance of whole Genome sequencing. (5 marks)

Ans: • Many bacteria and fungi are important as producers of valuable products,

- for example antibiotics, medicines,
- vitamins, soy sauce, cheese, yoghurt, and other foods made from milk.

There is considerable interest in analyzing the genomes of these (pu¨hler and selbitschka 2003). Other bacteria are valuable genomic models because of their capacity to degrade environmental pollutants;

- For example,
- the marine bacterium Alcanivorax borkumensis is a genomic model because it produces surfactants and is associated with the biodegradation of hydrocarbons in oil spills (ro¨ling et al. 2004).

Evolutionary position

- Whole-genome analysis of organisms at crucial or disputed positions in the tree of life can be expected to contribute significantly to our knowledge of evolution.
- The sea squirt, ci. Intestinalis, was chosen as a model because it belongs to a group, the Urochordata, with properties similar to the ancestors of vertebrates.
- The study of this species should provide valuable information about the early evolution of the phylum to which we belong ourselves.

- Many other organisms, although not on the list for a genome project to date, have a strong case for being declared as model species for evolutionary arguments. These include the velvet worm, Peripatus, traditionally seen as a missing link between the arthropods and annelids, but now classified as a separate phylum in the panarthropoda lineage (nielsen 1995), and the springtail, folsomia candida, formerly regarded as a primitive insect, but now sug-gested to have developed the hexapod bodyplan before the insects separated from the crustaceans (nardi et al. 2003).

Comparative purposes

- Over the last few years, genomicists have realized that assigning functions to genes and recognizing promoter sequences in a model genome can greatly benefit from comparison with a set of carefully chosen reference organisms at defined phylogenetic distances.
- Comparative genomics is developing an increasing array of bioinformatics techniques, such as
- synteny analysis,
- phylogenetic footprinting,
- and phylogenetic shadowing
- by which it is possible to understand aspects of a model genome from other genomes.
- One of the main reasons for sequencing the chimpanzee's Genome was to illuminate the human genome, and a variety of fungi were sequenced to illuminate the genome of S. Cerevisae.

Ecological significance

- It will be clear that ecological arguments have only played a minor role in the selection of species for whole-genome sequencing, but we expect them to become more important in the future

60. ecological genomics

Biology and Informational Science

- ✓ The HGP changed how we view & practice biology.
- ✓ Biology is an informational science.

✓ Digital genome

✓ Environmental signals Biology has become a cross-disciplinary science.

Bioinformatics as an intersecting discipline

61. Importance of biodiversity .5 marks

The new range of models should embrace diverse phylogenetic lineages, varying in their physiology and life-history strategy.

For example, the model plants arabidopsis and rice both employ the C3 photosynthetic pathway. To complement our genomic knowledge of primary production, new models should be chosen among plants utilizing C4 photosynthesis or crassulacean acid metabolism (CAM). Considering the diversity of life histories, species differing in their mode of reproduction and dispersal capacity should be chosen; for example, hermaphoditism versus gonochorism, parthenogenesis versus bisexual reproduction, et

SUBJECTIVE

2 Marks Questions

1. DEFINE COMMUNITY STRUCTURE?

Community structure is essentially the composition of a community, including the number of species in that community and their relative numbers. It can also be interpreted more broadly, to include all of the patterns of interaction between these different Species.

2. WHAT IS INTERMEDIATE DISTRUBANCE HYPOTHESIS?

The effects of human disturbance fall within the framework of the intermediate disturbance hypothesis. Though intense human disturbance reduces species diversity, moderate levels of disturbance may increase the diversity of some communities such as the European chalk grasslands

3. WHAT IS ECOLOGICAL GENOMICS?

A scientific discipline that studies the structure and functioning of a genome with the aim of understanding the relationship between the organism and its biotic and abiotic environments

4. ENLIST FOUR REASONS FOR LACKING OF FOSSIL RECORD?

1. Soft tissues are rarely preserved
2. Movement of the earth's crust has destroyed and covered many fossils
3. Fossilization takes place only in certain types of habitats and favorable environments
4. Paleontologists have not dug up every place on earth

5. DEFINE GENOME

A genome is an organism's complete set of DNA, including all of its genes. Each genome contains all of the information needed to build and maintain that organism.

6. DEFINE BIOINFORMATICS?

Bioinformatics is an interdisciplinary field that develops methods and software tools for understanding biological data. As an interdisciplinary field of science, bioinformatics combines biology, computer science, information engineering, mathematics and statistics to analyze and interpret biological data.

7. WHAT DOES INTERMEDIATE DISTURBANCE HYPOTHESIS SUGGEST?

The intermediate disturbance hypothesis suggests that communities with a medium (intermediate) level of disturbance may have greater species diversity than communities with very frequent or very rare disturbances.

8. ROLE OF PRIMERS IN PCR?

A primer is a short single strand of RNA or DNA (generally about 18-22 bases) that serves as a starting point for DNA synthesis. It is required for DNA replication because the enzymes that catalyze this process, DNA polymerases, can only add new nucleotides to an existing strand of DNA in the PCR.

9. PROTEOMICS DEFINITION?

Proteomics is the study of all the proteins in the cell. Mass spectrometry may also be used to determine the amino acid sequence of a protein. For this application, the protein is cleaved with a protease.

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10. ENLIST TYPES OF SERVERS USED IN BIOINFORMATICS?

1. WWW servers
2. Database servers
3. Intensive computing servers

11. ENLIST THREE MAJOR TECHNOLOGICAL DEVELOPMENTS OF 1990S?

1. Micro technology
2. Computing
3. Communication

12. HOW SEQUENCE ASSEMBLY IS DONE USING SHOTGUN APPROACH?

Master sequence ← short sequences, simply by examining the sequences for overlaps. No need any prior knowledge of the genome

13. WHAT ARE THREE LAYERS OF GENOME ANNOTATION?

1. Nucleotide-level annotation
2. Protein level annotation
3. Process level annotation

14. DIFFERENTIATE BETWEEN SPECIES TREE AND GENE TREE.

SPECIES TREE	GENE TREE
Contains only one representative from each species	Often contains a number of genes from a single species
All nodes indicate speciation events	Nodes relate either to speciation or gene duplication events.

15. DEFINE MICRO ECOLOGY?

The possibility of working with molecules on the scale of a few micrometers, given by advances in laser technology, has been very important for one of genomics' most conspicuous achievements, the development of the gene chip.

16. WHAT IS THE HIDDEN MARKOV MODEL

It may generally be used in pattern recognition problems, anywhere there may be a model producing a sequence of observations. In bioinformatics, it has been used in sequence alignment, in silico gene detection, structure prediction, data-mining literature, and so on.

3 MARKS QUESTIONS

1. EXPLAIN SPECIES RICHNESS AND SPECIES DIVERSITY WITH EXAMPLE?

SPECIES RICHNESS is the number of different species in a particular community. **For example** if we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first. Communities with the highest species richness tend to be found in areas near the equator and Communities with the lowest species richness lie near the poles

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SPECIES DIVERSITY is a measure of community complexity. It is a function of both the number of different species in the community (species richness) and their relative abundances (species evenness. **For example:** • A forest community with 20 different kinds of trees would have greater species diversity than a forest community with only 5 kinds of trees.

2. HOW THE EVOLUTIONARY HISTORY CONVERTS IN TO DOCUMENT?

- Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history
- DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago
- mtDNA evolves rapidly and can be used to explore recent evolutionary events

3. NAME THE PHYLA IN AMOEBOID PROTOZOANS? WRITE THE NAME OF THE PHYLA WHICH INCLUDES AMOEBOID PROTOZOANS AND DESCRIBE THEIR LOCOMOTION?

Phylum: - Sarcodinids (Amoeboid protozoan)

Contains 4 phyla: Foraminifera, Radiozoa, Amoebozoa, and Heliozoa

Locomotion:- locomotion occurs due to the streaming of the cytoplasm, that results into the formation of the pseudopodia "False feet", and it helps amoeba for the locomotion.

Other Characteristics

- Most reproduce asexually through binary fission.
- Food is usually captured by phagocytosis.
- Body types range from free flowing to rigid with skeletal supports

4. INTERACTION B/W GENES WITHIN GENOME?

The interactions between the genes within the genome and the dynamic character of the genome on an evolutionary scale have been sketched brilliantly by Dover (1999) as an internal tangled bank. This idea goes back to Darwin (1859) who, after investigating the banks of hollow roads in the English countryside, was attracted by the great variety of organisms tangled together.

Darwin considered the way in which all organisms depended on each other as the template for evolution. Inspired by Darwin, Dover (1999) made a distinction between the 'external tangled bank' (the ecology) and the 'internal tangled bank' (the genome), attributing to them complementary roles in the evolutionary process

5. HOW KEYSTONE SPECIES DIFFER FROM FOUNDATION SPECIES?

- A keystone species is a species that has a extremely large effect on community structure relative to its biomass or abundance.
- Keystone species differ from foundation species in two main ways
  i. they are more likely to belong to higher trophic levels (to be top predators),
  ii. they act in more diverse ways than foundation species, which tend to modify their environment

6. EXPLAIN SEQUENCING?

Sequencing means to make DNA/Protein short fragments into in an order.

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Protein sequencing

The laborious edman degradation technique has now been replaced mostly by sequencing using mass spectrometry, in which masses are estimated for peptides generated from proteolytic digests of the protein, while the fragment pattern obtained is compared with entries in a database that include sizes predicted from the genomic sequence.

7. WRITE AT LEAST THREE REASONS TO USE PHYLOGENETIC.

1. **Understand the lineage of different species:**
2. **Understand how various functions evolved:**
3. **Perform multiple sequence alignment**
4. **Predict gene function (phylogenetic footprint)**

5 MARKS Questions

1. EXPLAIN IMPORTANCE OF BIODIVERSITY?

Biodiversity has a number of functions on the Earth.

These are as follows:

- Maintaining balance of the ecosystem:
- Recycling and storage of nutrients,
- combating pollution, and stabilizing climate, protecting water resources,
- Forming and protecting soil and maintaining Eco-balance.
- Intrinsic value
- A philosophical concept of the value of something independent of its value to anyone or anything

Provision of biological resources

- Provision of medicines and pharmaceuticals, food for the human population and animals, ornamental plants, wood products, breeding stock and diversity of species, ecosystems and genes.
- Quinone is a drug used for the treatment of malaria which is obtained from the Cinchona genus of plants.

Social benefits:- Recreation and tourism, cultural value and education and research. The role of biodiversity in the following areas will help make clear the importance of biodiversity in human life:

Biodiversity and food: - 80% of human food supply comes from 20 kinds of plants. But humans use 40,000 species for food, clothing and shelter. Biodiversity provides for variety of foods for the planet.

Biodiversity and human health:- The shortage of drinking water is expected to create a major global crisis. Biodiversity also plays an important role in drug discovery and medicinal resources. Medicines from nature account for usage by 80% of the world's population.

Biodiversity and industry:- Biological sources provide many industrial materials. These include fiber, oil, dyes, rubber, water, timber, paper and food.

Biodiversity and culture: - Biodiversity enhances recreational activities like bird watching, fishing, trekking etc. It inspires musicians and artists.

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2. WHAT IS EVOLUTIONARY SIGNIFICANCE OF SEQUENCING GENOME IN ECOLOGY; EXPLAIN WITH EXAMPLES.

Ecological significance

It will be clear that ecological arguments have only played a minor role in the selection of species for whole-genome sequencing, but we expect them to become more important in the future. The new range of models should embrace diverse phylogenetic lineages, varying in their physiology and life-history strategy. **For example**, the model plants arabidopsis and rice both employ the C3 photosynthetic pathway. To complement our genomic knowledge of primary production, new models should be chosen among plants utilizing C4 photosynthesis or crassulacean acid metabolism (CAM). Considering the diversity of life histories, species differing in their mode of reproduction and dispersal capacity should be chosen; **for example**, hermaphoditism versus gonochorism, parthenogenesis versus bisexual reproduction, etc

3. What are phylogenetic trees (how genes are related)

- Phylogentics tree or evolutionary tree is a diagram or branching tree.
- Which shows the evolutionary relationship between the different species depending upon their phylogeny based upon similarities and differences in their physical and genetic characteristics
- All the life on earth is part of the single phylogenic tree showing only one common ancestor and all species have been evolved from that common ancestor.

Reasons for using the phylogenetic trees

- Understanding the linage of the different species
- Understanding how the various functions evolved among different species
- Perform multiple sequence alignment
- Predict gene functions (phylogenetic footprints)

How genes are related

- Gene tree contains no of genes from a single species
- Every node of the gene tree shows the speciation or gene duplications events.
- Gene families are groups of related genes that share a common ancestor.
- Paralogs are genes with similar sequences from within the same species
- orthologs are genes with similar sequences in different species

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4. WRITE FIVE FACTORS THAT AFFECTS COMMUNITY STRUCTURE ?

The structure of a community is the result of many interacting factors, both abiotic (non-living) and biotic (living organism-related).

Here are some important factors that influence community structure:

- The climate patterns of the community's location.
- The geography of the community's location.
- The heterogeneity (patchiness) of the environment
- The frequency of disturbances, or disruptive events.
- Interactions between organisms.

5. WHY THE STUDY OF WHOLE GENOME IS IMPORTANT FOR THE EVOLUTIONARY STUDY?

- Whole-genome analysis of organisms at crucial or disputed positions in the tree of life can be expected to contribute significantly to our knowledge of evolution.
- The sea squirt, **ci. Intestinalis**, was chosen as a model because it belongs to a group, the Urochordata, with properties similar to the ancestors of vertebrates.
- The study of this species should provide valuable information about the early evolution of the phylum to which we belong ourselves.
- Many other organisms, although not on the list for a genome project to date, have a strong case for being declared as model species for evolutionary arguments. These include the velvet worm, Peripatus, traditionally seen as a missing link between the arthropods and annelids, but now classified as a separate phylum in the panarthropoda lineage , and the springtail, folsomia candida, formerly regarded as a primitive insect, but now suggested to have developed the hexapod body plann before the insects separated from the crustaceans.

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